PROJECT DESIGN PHASE

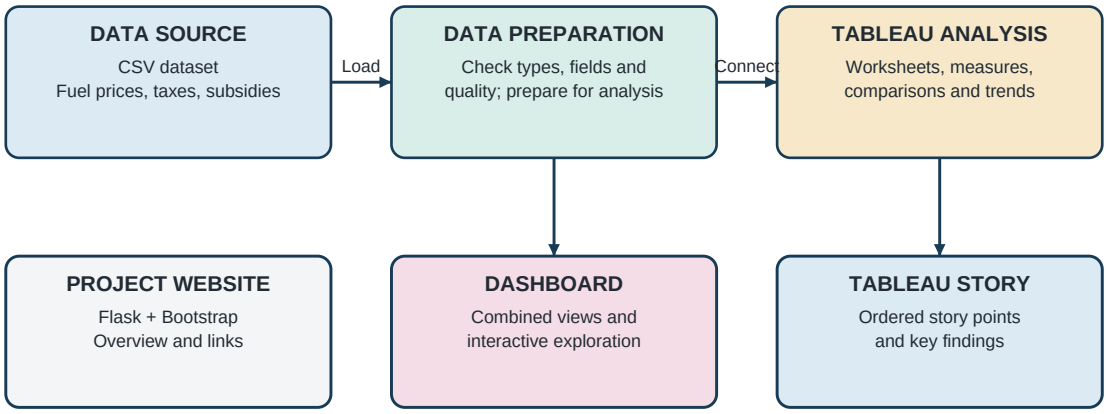
Solution Architecture

Date	19 September 2026
Team ID	
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Analysis
Maximum Marks	4 Marks

Solution Architecture

The architecture connects the project dataset to data preparation and Tableau analysis, then presents results through an interactive dashboard and a Tableau Story. A supporting Flask website can introduce the project and provide links to published outputs. The design is suitable for a descriptive analytics project and does not require a separate data warehouse.

Solution Architecture Diagram



Architecture reflects the implemented Tableau analysis and supporting website; no Snowflake database is assumed.

Component Description Table

Component	Description / Responsibility
Data source	CSV file containing date, country, region, income level, subsidy level, petrol/diesel/LPG prices, Brent crude price and tax percentage.
Data preparation	Review field names and data types, check for missing/duplicate records, and ensure measures and dates are appropriate for analysis. Dataset review reported 27,468 records, 84 countries and no missing or exact duplicate rows.
Visualization layer	Tableau worksheets show country comparisons, fuel-type differences, tax/subsidy patterns, time trends and Brent crude price trends.
Dashboard layer	Combines key visualizations in one place for comparison and exploration. Labels, legends and units should be clear.
Story layer	Organizes the analysis into a sequence of story points to explain the main patterns and insights.
Presentation layer	Tableau Public shares the published dashboard and Story; the Flask/Bootstrap website provides project context and links.
Repository / documentation	GitHub stores project files and supports project sharing. The dataset and published outputs should be referenced in the final report.

Data flow summary: CSV dataset → Tableau data connection and preparation → analytical worksheets → dashboard and Story → Tableau Public / supporting project website.

Design note: This diagram describes the actual project components at a high level. It does not imply that Snowflake or a separate SQL database was used.